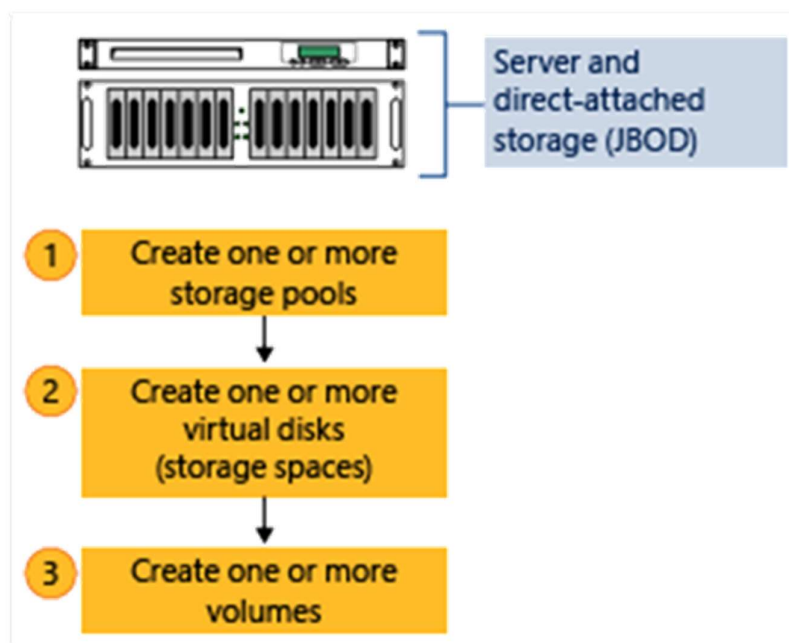


Deploy Storage Spaces on a stand-alone server

To create a storage space, you must first create one or more storage pools. A storage pool is a collection of physical disks. A storage pool enables storage aggregation, elastic capacity expansion, and delegated administration.

A storage space appears to the Windows operating system as a regular disk from which you can create formatted volumes. When you create a virtual disk through the File and Storage Services user interface, you can configure the resiliency type (simple, mirror, or parity), the provisioning type (thin or fixed), and the size.

The following figure illustrates the Storage Spaces workflow.



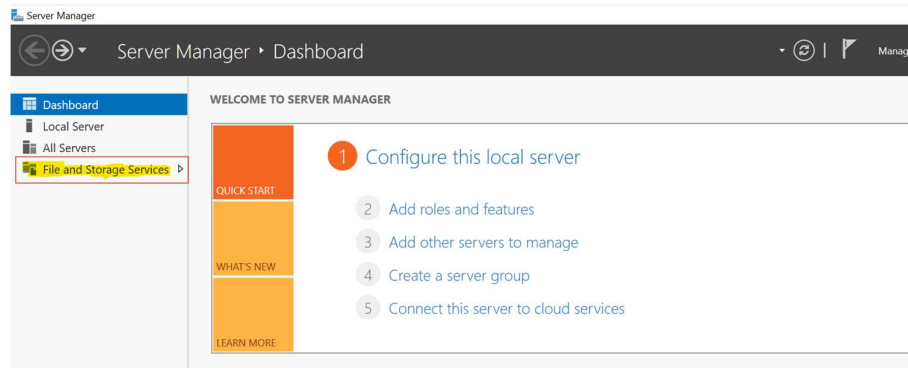
Resiliency type (simple, mirror, or parity)

Resiliency type	Disk requirements	When to use
Simple - Stripes data across physical disks - Maximizes disk capacity and increases throughput - No resiliency (doesn't protect from disk failure)	Requires at least one physical disk.	Don't use to host irreplaceable data. Simple spaces don't protect against disk failure. Use to host temporary or easily recreated data at a reduced cost. Suited for high-performance workloads where resiliency isn't required or is already provided by the application.
Mirror - Stores two or three copies of the data across the set of physical disks - Increases reliability but reduces capacity. Duplication occurs with every write. A mirror space also stripes the data across multiple physical drives. - Greater data throughput and lower access latency than parity - Uses dirty region tracking (DRT) to track modifications to the disks in the pool. When the system resumes from an unplanned shutdown and the spaces are brought back online, DRT makes disks in the pool consistent with each other.	Requires at least two physical disks to protect from single disk failure. Requires at least five physical disks to protect from two simultaneous disk failures.	Use for most deployments. For example, mirror spaces are suited for a general-purpose file share or a virtual hard disk (VHD) library.
Parity - Stripes data and parity information across physical disks - Increases reliability when it's compared to a simple space, but somewhat reduces capacity - Increases resiliency through journaling. This function helps prevent data corruption if an unplanned shutdown occurs.	Requires at least three physical disks to protect from single disk failure.	Use for workloads that are highly sequential, such as archive or backup.

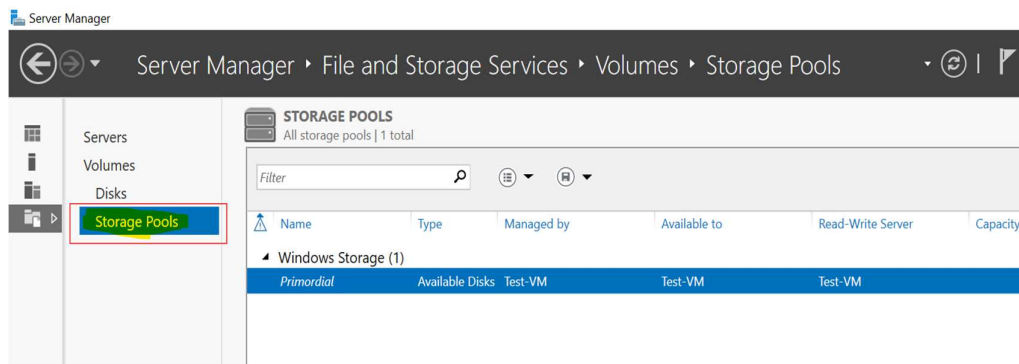
Step 1: Create a storage pool

You must first group available physical disks into one or more storage pools.

1. In the Server Manager navigation pane, select **File and Storage Services**.

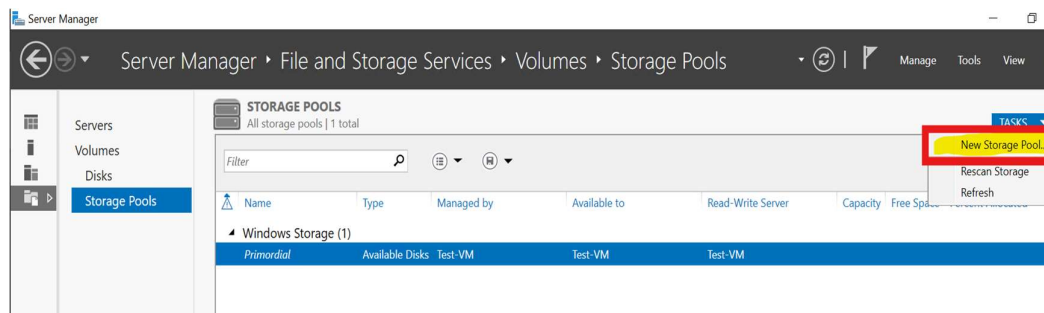


2. Under **Volumes**, select **Storage Pools**.

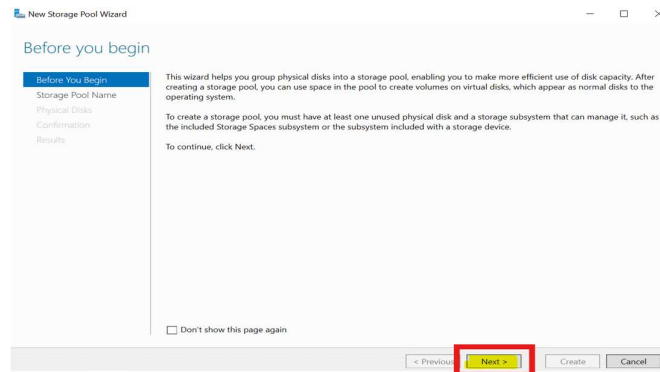


By default, available disks are included in a pool that is named the *primordial* pool. If no primordial pool is listed under STORAGE POOLS, this situation indicates that the storage doesn't meet the requirements for Storage Spaces.

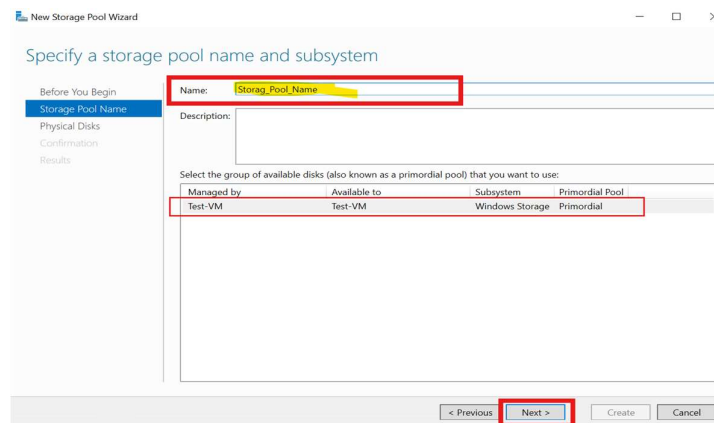
3. Under **STORAGE POOLS**, select the **TASKS** list, and then select **New Storage Pool**. The New Storage Pool Wizard opens.



4. On the **Before you begin** page, select **Next**.

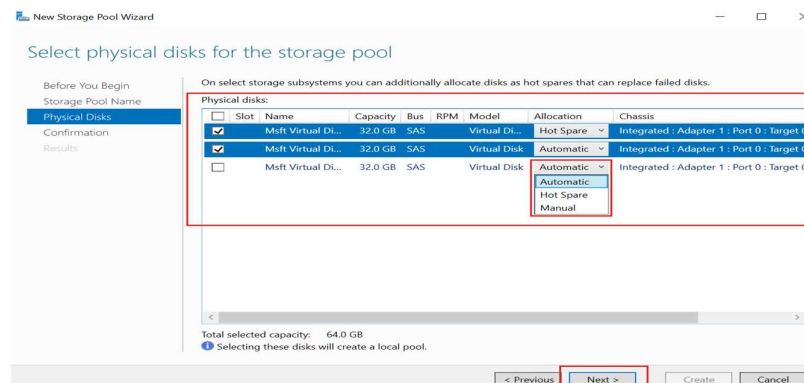


5. On the **Specify a storage pool name and subsystem** page, enter a name and optional description for the storage pool, select the group of available physical disks that you want to use, and then select **Next**.

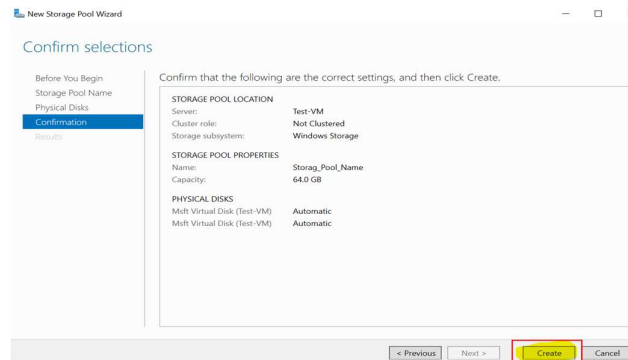


6. On the **Select physical disks for the storage pool** page, do the following, and then select **Next**:

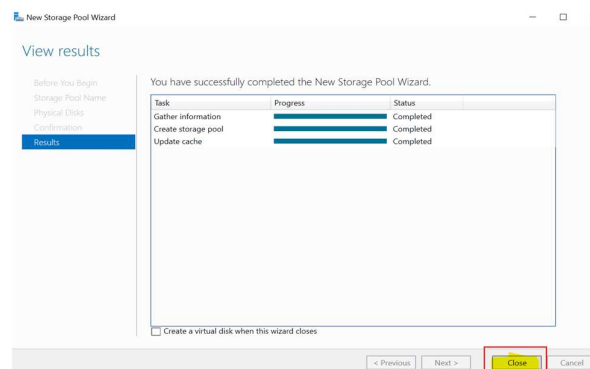
- Select the check box next to each physical disk that you want to include in the storage pool.
- If you want to designate one or more disks as hot spares, under **Allocation**, select the drop-down arrow, then select **Hot Spare**.



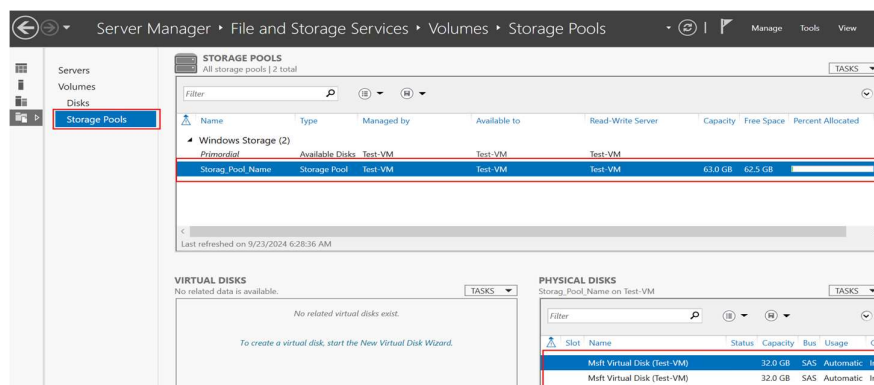
7. On the **Confirm selections** page, verify that the settings are correct, and then select **Create**.



8. On the **View results** page, verify that all tasks completed, and then select **Close**.



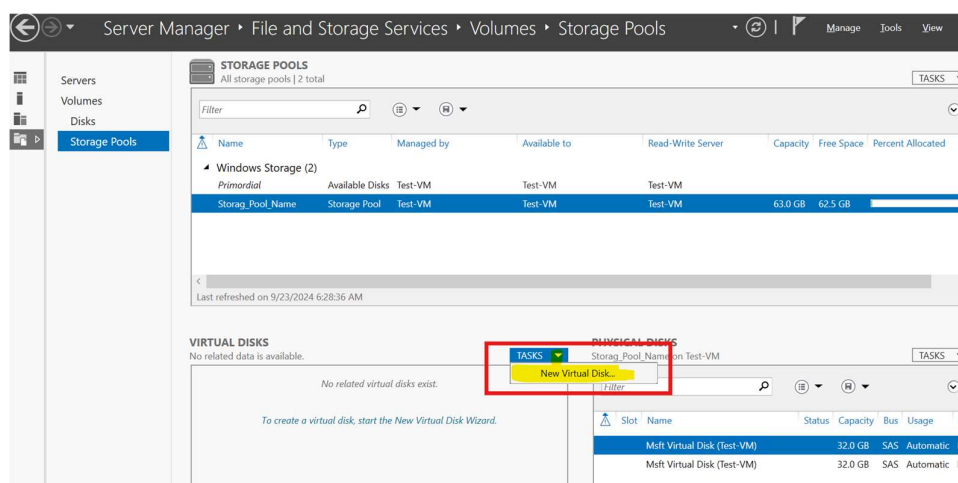
9. Under **STORAGE POOLS**, verify that the new storage pool is listed.



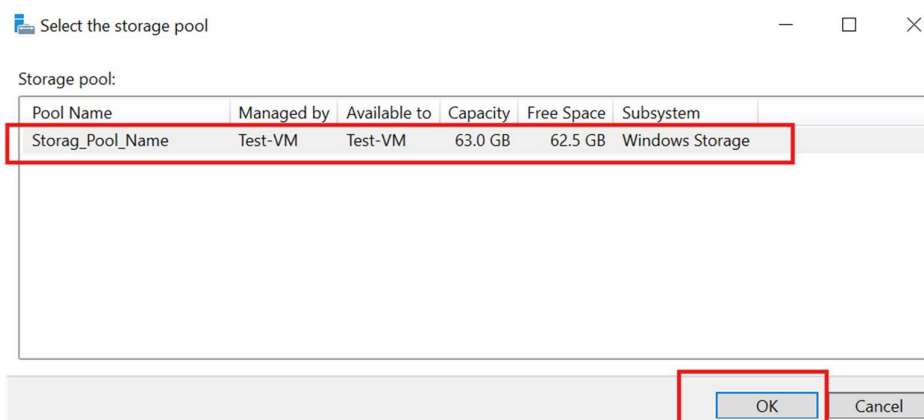
Step 2: Create a virtual disk

Next, you must create one or more virtual disks from the storage pool. When you create a virtual disk, you can select how the data is laid out across the physical disks. This selection affects both reliability and performance. You can also select whether to create thin- or fixed-provisioned disks.

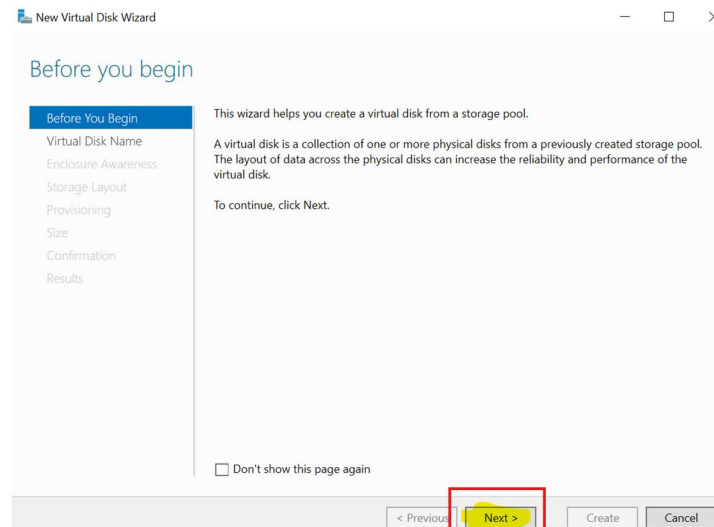
1. If the New Virtual Disk Wizard isn't open yet, on the **Storage Pools** page in Server Manager, under **STORAGE POOLS**, make sure that the desired storage pool is selected.
2. Under **VIRTUAL DISKS**, select the **TASKS** list, and then select **New Virtual Disk**. The New Virtual Disk Wizard opens.



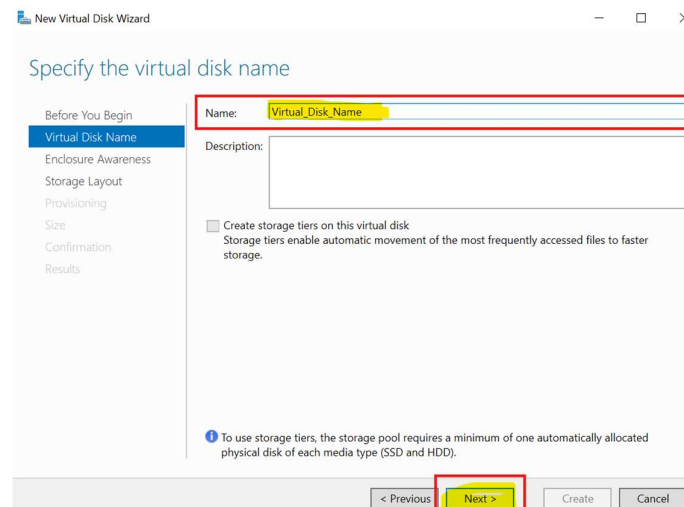
3. On the **Select the storage pool** page, select the desired storage pool, and then select **Next**.



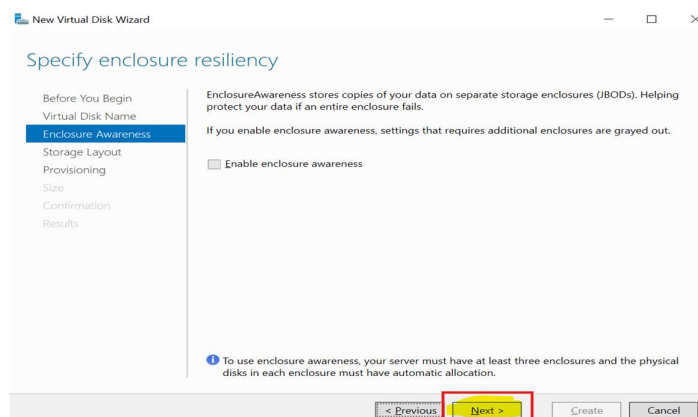
4. On the **Before you begin** page, select **Next**.



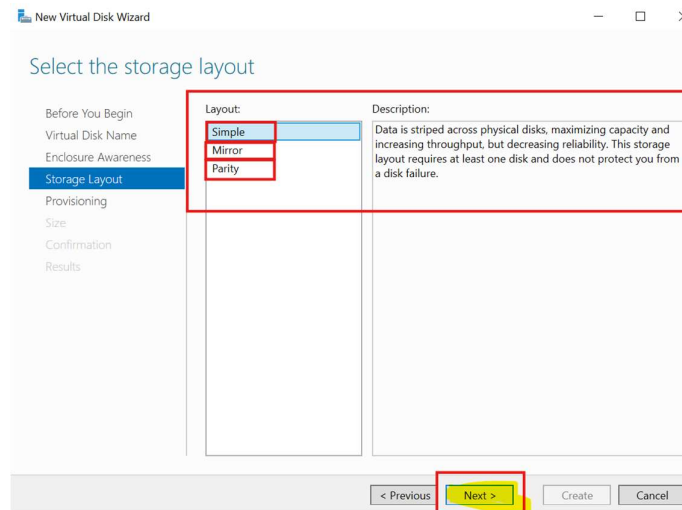
5. On the **Specify the virtual disk name** page, enter a name and optional description, then select **Next**.



6. On the **Specify enclosure resiliency** page, then select **Next**.



7. On the **Select the storage layout** page, select the desired layout, then select **Next**.



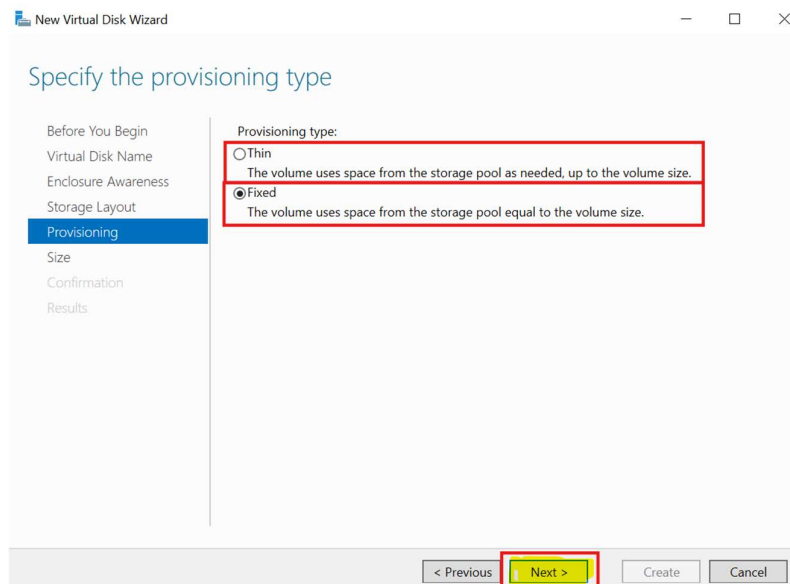
8. On the **Specify the provisioning type** page, select one of the following options, then select **Next**.

- **Thin**

With thin provisioning, space is allocated on an as-needed basis. This selection optimizes the use of available storage. However, because this setting enables you to over-allocate storage, you must carefully monitor how much disk space is available.

- **Fixed**

With fixed provisioning, the storage capacity is allocated immediately, at the time, a virtual disk is created. Therefore, fixed provisioning uses space from the storage pool that is equal to the virtual disk size.



9. On the **Specify the size of the virtual disk** page, perform one of the following actions:

- If you selected thin provisioning in the previous step, follow these steps:
 - a. In the **Virtual disk size** box, enter a virtual disk size.
 - b. Select the units (**MB**, **GB**, or **TB**), and then select **Next**.
- If you selected fixed provisioning in the previous step, select one of the following options:
 - **Specify size**

To specify a size, enter a value in the **Virtual disk size** box, then select the units (**MB**, **GB**, or **TB**).

- **Maximum size**

Select this option to create a virtual disk that uses the maximum capacity of the storage pool.

The screenshot shows the 'New Virtual Disk Wizard' window with the title bar 'New Virtual Disk Wizard'. The main heading is 'Specify the size of the virtual disk'. On the left, a sidebar lists the steps: 'Before You Begin', 'Virtual Disk Name', 'Enclosure Awareness', 'Storage Layout', 'Provisioning', 'Size' (highlighted in blue), 'Confirmation', and 'Results'. The main area shows 'Free space in this storage pool: 62.0 GB'. There are two radio button options: 'Specify size:' (selected) and 'Maximum size'. The 'Specify size:' option has a text input field containing '62.0' and a dropdown menu set to 'GB'. The 'Maximum size' option is unselected. At the bottom, there is a blue information icon and the text 'The virtual disk might take additional space to create a write-back cache.' The bottom navigation bar contains four buttons: '< Previous', 'Next >' (highlighted in yellow), 'Create', and 'Cancel'.

10. On the **Confirm selections** page, verify that the settings are correct, and then select **Create**.

New Virtual Disk Wizard

Confirm selections

Before You Begin
Virtual Disk Name
Enclosure Awareness
Storage Layout
Provisioning
Size
Confirmation
Results

Confirm that the following are the correct settings, and then click Create.

VIRTUAL DISK LOCATION

Server: Test-VM
Subsystem: Windows Storage
Storage pool name: Storag_Pool_Name
Status: OK
Free space: 62.5 GB

VIRTUAL DISK PROPERTIES

Name: Virtual_Disk_Name
Storage tiers: Disabled
Storage layout: Simple
Provisioning type: Fixed
Total requested size: 62.0 GB
Enclosure awareness: None

< Previous Next > **Create** Cancel

11. On the **View results** page, verify that all tasks are completed, and then select **Close**.

New Virtual Disk Wizard

View results

Before You Begin
Virtual Disk Name
Enclosure Awareness
Storage Layout
Provisioning
Size
Confirmation
Results

The New Virtual Disk Wizard successfully completed.

Task	Progress	Status
Gather information		Completed
Create virtual disk		Completed
Rescan disks		Completed
Initialize disk		Completed
Update cache		Completed

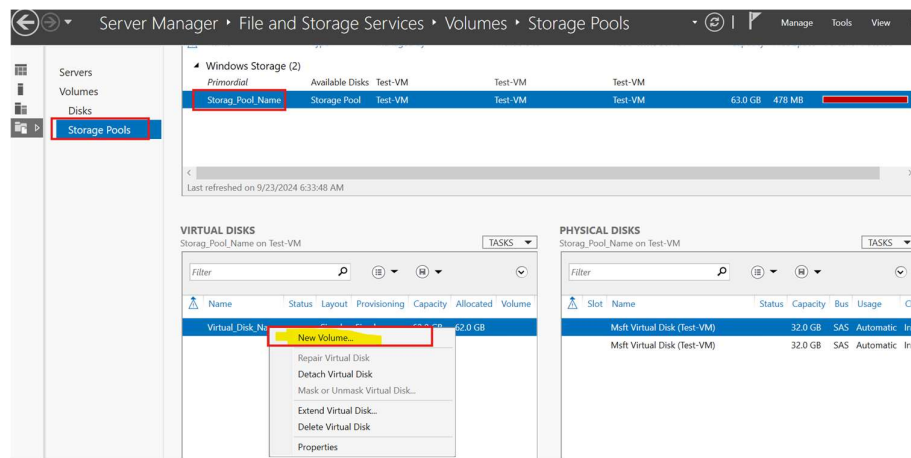
☐ Create a volume when this wizard closes

< Previous Next > **Close** Cancel

Step 3: Create a volume

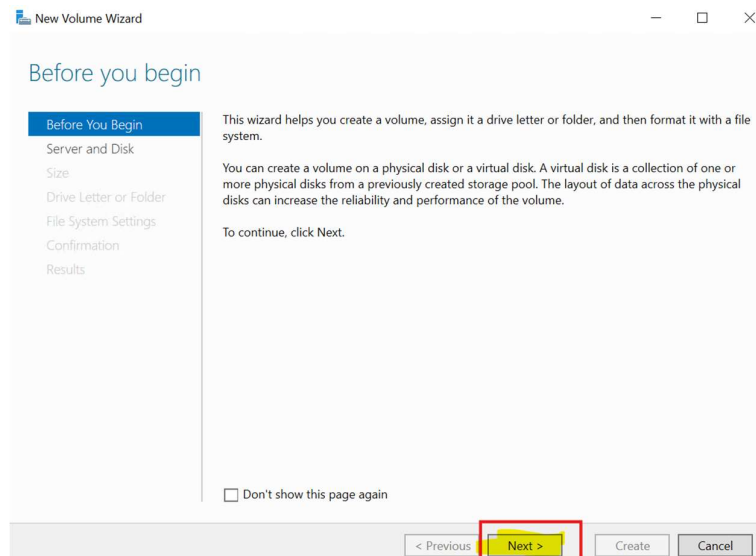
Next, you must create a volume from the virtual disk. You can assign an optional drive letter or folder, then format the volume with a file system.

1. If the New Volume Wizard isn't open yet, on the **Storage Pools** page in Server Manager, under **VIRTUAL DISKS**, right-click the desired virtual disk, and then select **New Volume**.



The New Volume Wizard opens.

2. On the **Before you begin** page, select **Next**.



3. On the **Select the server and disk** page, do the following, and then select **Next**.
 - a. In the **Server** area, select the server on which you want to provision the volume.
 - b. In the **Disk** area, select the virtual disk on which you want to create the volume.

New Volume Wizard

Select the server and disk

Before You Begin

- Server and Disk
- Size
- Drive Letter or Folder
- File System Settings
- Confirmation
- Results

Server:

Provision to	Status	Cluster Role	Destination
Test-VM	Online	Not Clustered	Local

Refresh Rescan

Disk:

Disk	Virtual Disk	Capacity	Free Space	Subsystem
Disk 2		32.0 GB	32.0 GB	
Disk 5	Virtual_Disk_Name	62.0 GB	62.0 GB	Windows Storage

< Previous Next > Create Cancel

4. On the **Specify the size of the volume** page, enter a volume size, specify the units (**MB**, **GB**, or **TB**), and then select **Next**.

New Volume Wizard

Specify the size of the volume

Before You Begin

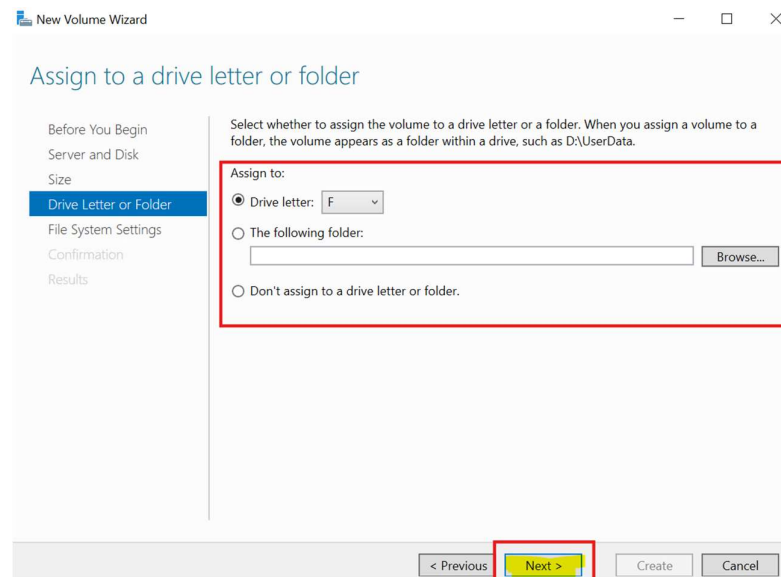
- Server and Disk
- Size
- Drive Letter or Folder
- File System Settings
- Confirmation
- Results

Available Capacity: 62.0 GB
Minimum size: 8.00 MB

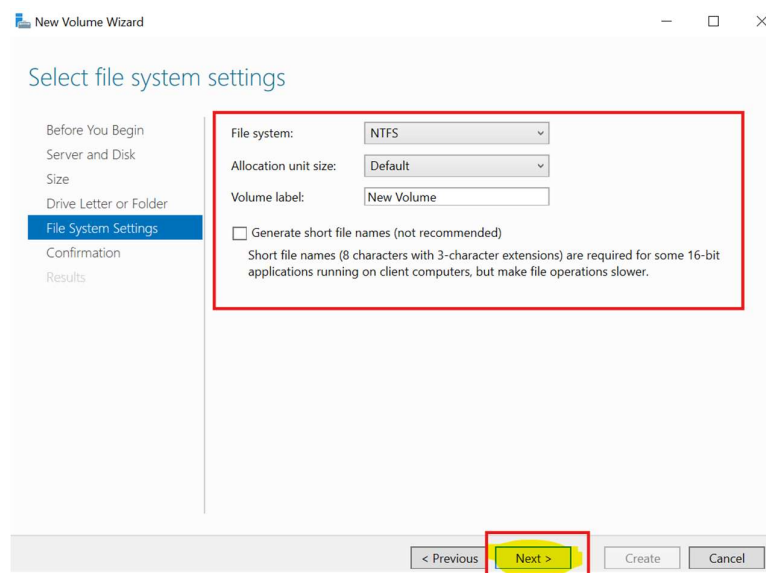
Volume size: 62.0 GB

< Previous Next > Create Cancel

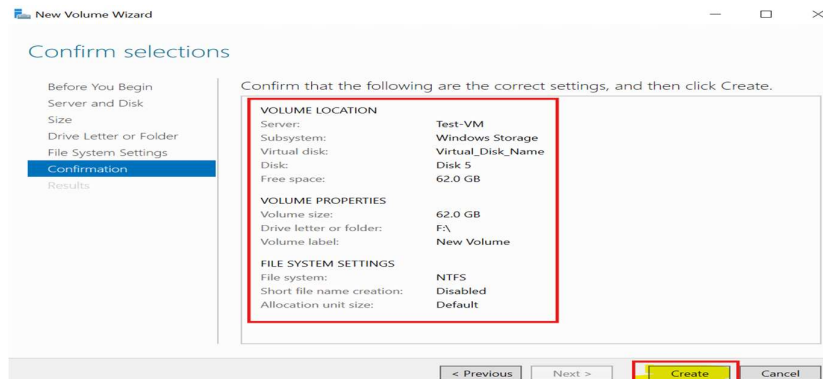
5. On the **Assign to a drive letter or folder** page, configure the desired option, and then select **Next**.



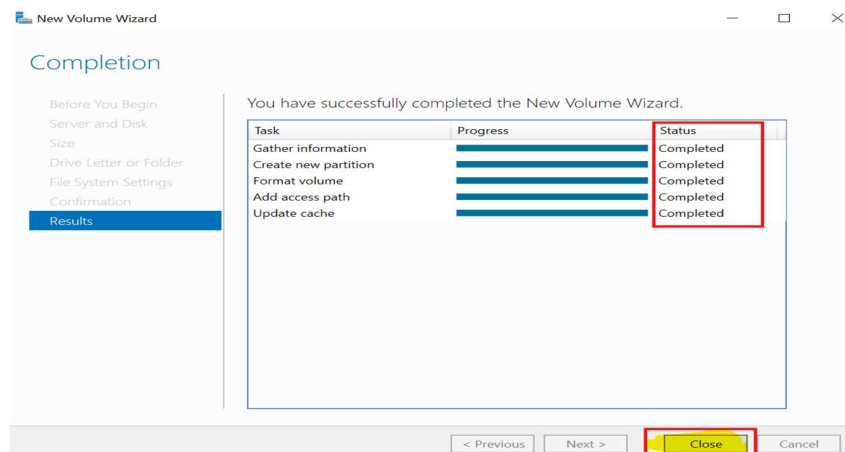
6. On the **Select file system settings** page, do the following, and then select **Next**.
- In the **File system** list, select either **NTFS** or **ReFS**.
 - In the **Allocation unit size** list, either leave the setting at **Default** or set the allocation unit size.
 - Optionally, in the **Volume label** box, enter a volume label name, for example **HR Data**.



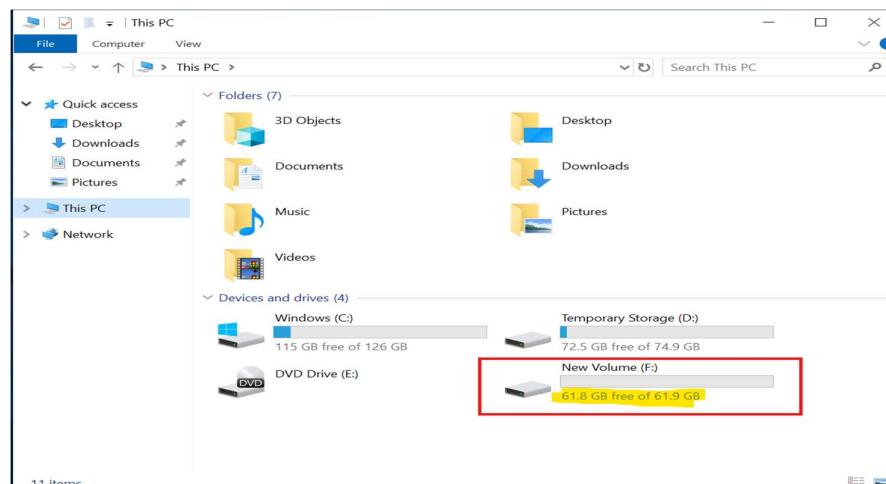
7. On the **Confirm selections** page, verify that the settings are correct, and then select **Create**.



8. On the **View results** page, verify that all tasks are completed, and then select **Close**.

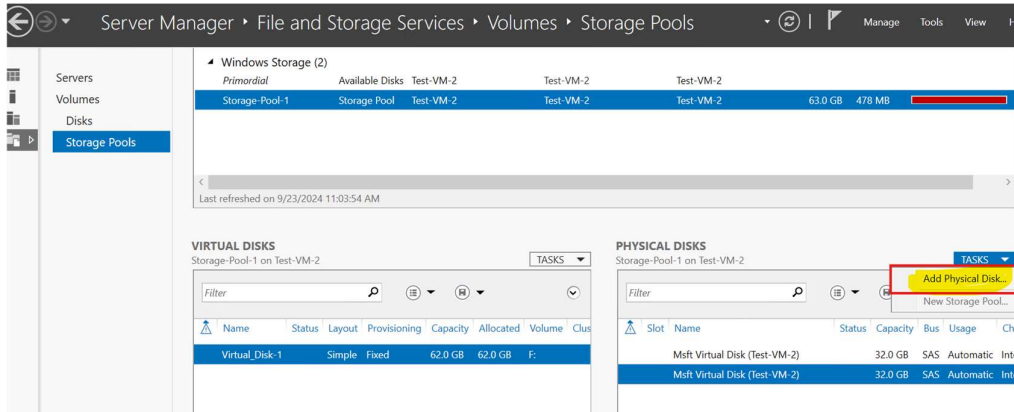


9. To verify that the volume was created, in Server Manager, select the **Volumes** page. The volume is listed under the server where it was created.

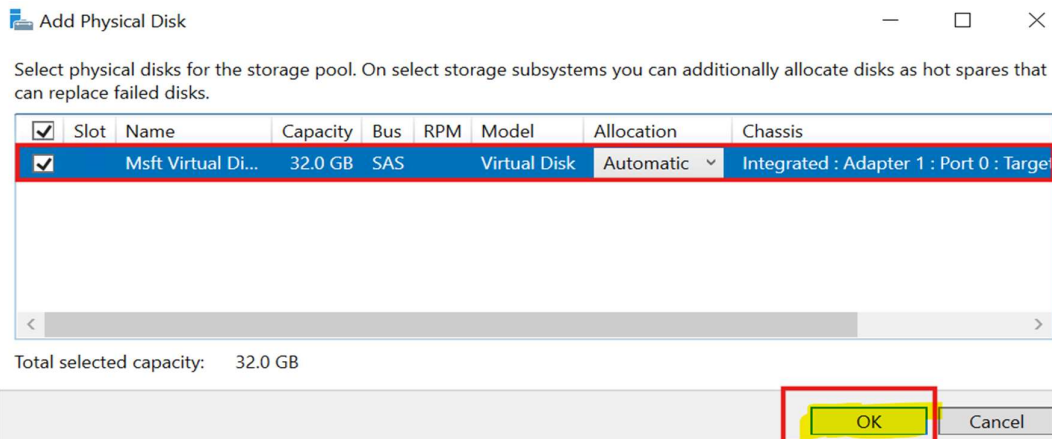


FOR ADD EXTRA DISK IN EXITING STORAGE POOL

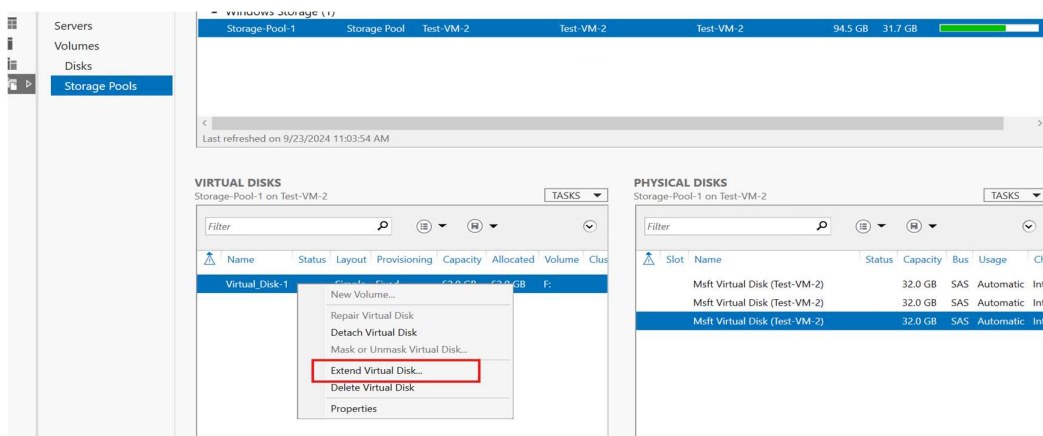
1. Under **PHYSICAL DISKS**, select the **TASKS** list, and then select **ADD Physical Disk**. The New Wizard will open



2. The wizard will show the disk list Available Disk for Add, Select the disk from the List and click ok.



3. Under **VIRTUAL DISKS**, right click on the exiting **Volume disk** and choose **Extend Virtual Disk**.



4. On the **Specify the size of the volume** page, enter new volume size, specify the units (**MB**, **GB**, or **TB**), and then select **Ok**.

Extend Virtual Disk

Maximum size: 126 GB

Current size: 60.0 GB

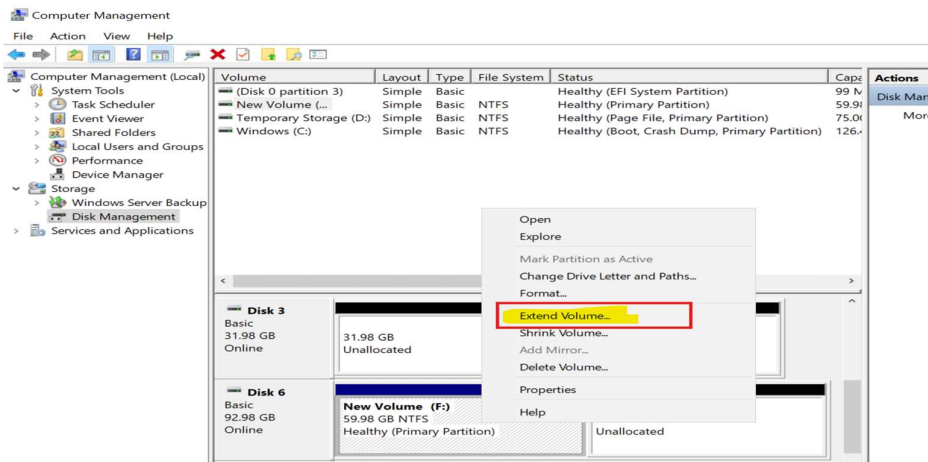
☒ Specify size:

126 GB

☐ Maximum size

OK Cancel

5. After that go to the Compute management Portal Choose Disk Management then right click on Virtual Volume disk and choose Extent Volume.



6. The extend wizard will open and new added size will auto configure we need to click on next.

Extend Volume Wizard

Select Disks

You can use space on one or more disks to extend the volume.

Available:

Add >

< Remove

< Remove All

Selected:

Disk 6 67583 MB

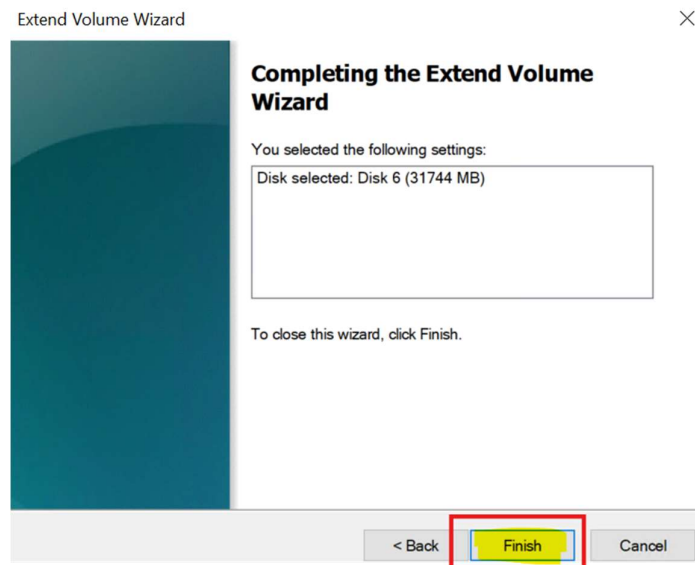
Total volume size in megabytes (MB): 129006

Maximum available space in MB: 67583

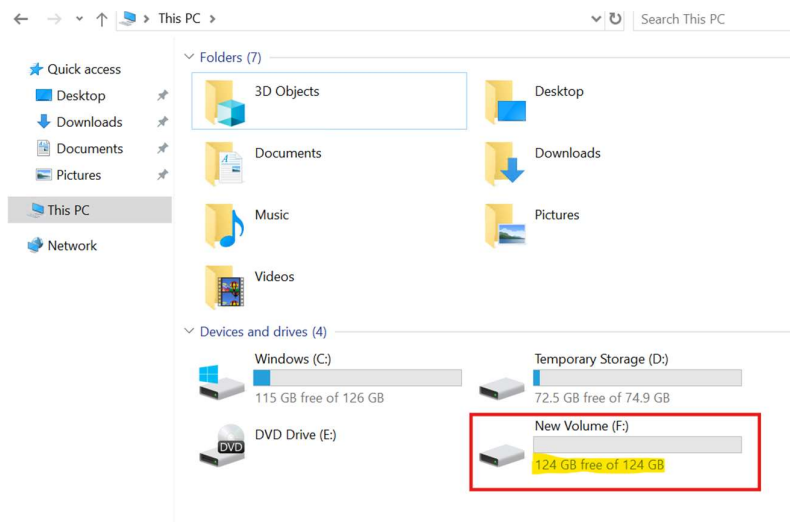
Select the amount of space in MB: 67583

< Back Next > Cancel

7. Click on finish to complete the Disk extend task.



8. Verify the new size is visible to Virtual volume.



NOTE: -

9. We can create Multiple Virtual Disks with one storage-Pool.
10. Once Data disk Added to Storage Pool can't be remove. Only Possible If we delete connected Virtual Disks.
11. After creation of storage pool if any data disk is unplugged from Portal and desired storage pool will be disconnected.
12. We cannot Add or create Storage Pool with OS disk.